

TAUSEEF IQBAL

Machine Learning Engineer | LLM & RAG Systems | AI Automation

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PROFESSIONAL SUMMARY

Machine Learning Engineer specializing in AI Automation, LLM-powered applications, and Retrieval-Augmented Generation (RAG) systems. Hands-on experience building production ML/NLP pipelines, automation workflows with n8n, and AI agents that connect LLMs (LangChain, LangGraph, Hugging Face) to real business processes. Currently improving taxonomy accuracy and field-mapping quality for a US-based e-commerce client, raising mapping accuracy from ~60% to over 80%. Proficient in Python, LangChain, LangGraph, n8n workflow automation, Hugging Face, and vector search (ChromaDB, hybrid retrieval, RRF). Skilled at shipping end-to-end AI and automation systems on free-tier and self-hosted infrastructure, from data preprocessing through deployment.

TECHNICAL SKILLS

Languages: Python, SQL, Dart, JavaScript

LLM & NLP: LangChain, LangGraph, Hugging Face Transformers, OpenAI API, Groq, RAG Systems, Hybrid Retrieval (RRF), HyDE, Prompt Engineering, AST-Aware Chunking

ML/DL Frameworks: TensorFlow, PyTorch, Keras, Scikit-learn

Data & Vector Search: ChromaDB, Pandas, NumPy, Matplotlib, Seaborn, Plotly

Automation & Backend: n8n, FastAPI, Streamlit, Gradio, REST APIs, Git, Docker

Deployment: Oracle Cloud (Always Free), Vercel, Railway, Self-Hosted

Mobile: Flutter, Firebase

PROFESSIONAL EXPERIENCE

Junior Machine Learning Engineer | *DCUBE Technologies (Pvt) Limited, NUST Islamabad* Aug 2025 – Present

- Improve taxonomy accuracy and field-mapping quality for a US-based client's product mapping tool, raising mapping accuracy from ~60% to over 80% by analyzing key columns against a reference taxonomy system.
- Develop an LLM-powered Promotion Optimization System using LangChain and ChromaDB for intelligent vector search and retail insight generation.
- Build an automated WhatsApp AI Agent using NLP and transformer models to handle conversational customer interactions.
- Create the Belami Asset Automation Tool with Streamlit to automate enterprise asset template processing and validation for vendor Excel files.
- Manage Akeneo product data pipelines and image processing workflows for e-commerce catalog operations.

Machine Learning Engineer Intern | *DCUBE Technologies (Pvt) Limited, NUST Islamabad* Apr 2025 – Aug 2025

- Engineered a retail promotion optimization system with a Gradio and Flutter UI for business insight visualization.
- Fine-tuned Hugging Face transformer models for personalized recommendations and uplift prediction.

KEY PROJECTS

CodeSage — Natural-Language Code Search (RAG) — *Python, LangChain, Hybrid Retrieval, HyDE, React*

- Built a production-style Retrieval-Augmented Generation system that answers natural-language questions about a codebase (e.g. "where is authentication handled?") and cites the exact file and line numbers as evidence.
- Implemented AST-aware code chunking, hybrid retrieval with Reciprocal Rank Fusion (RRF), and HyDE (Hypothetical Document Embeddings) to improve retrieval precision over naive vector search.
- Designed a Groq/Gemini LLM router to balance cost, latency, and quality across free-tier model providers; built a React frontend with a dark editor-style UI, deployed on Oracle Cloud Always Free.

LexMind AI — Multi-Agent Legal RAG Platform — *LangGraph, Groq (Llama 3.1 70B), ChromaDB, FastAPI, React*

- Built a multi-agent legal intelligence platform using LangGraph to orchestrate specialized agents over a ChromaDB vector store for legal document retrieval and analysis.

- Developed a FastAPI backend and a React/TypeScript/shadcn frontend, deployed on Vercel (frontend) and Railway (backend).

AI Property Assistant — Real Estate Chatbot — *n8n, Supabase, OpenAI GPT, OpenWeatherMap API*

- Built a stateful real estate chatbot with dynamic tool switching, city-based response caching, and live weather context, orchestrated entirely as an n8n workflow.
- Integrated Supabase for conversation state and OpenAI GPT for natural-language understanding, running on free-tier infrastructure.

Faceless Video Automation Pipeline — *n8n, Groq, Pollinations.ai, FFmpeg*

- Designed a 27-node, self-hosted n8n workflow that converts a form submission into a fully voiced, captioned short-form video, delivered to the requester by email — built entirely on free-tier and self-hosted tools.
- Automated script generation (Groq), image generation (Pollinations.ai), and video assembly (FFmpeg) into a single end-to-end pipeline.

Promotion Optimization System — *LangChain, ChromaDB, Hugging Face, Gradio, Flutter*

- Built an intelligent retail promotion platform combining LLM-powered insights, uplift prediction, and vector search for product matching, with fine-tuned transformer models.

EDUCATION

Bachelor of Science in Computer Science

COMSATS University Islamabad, Abbottabad Campus | 2021 – 2025

Thesis: "Facial Keypoint Detection – Unveiling Faces" | Specialization: AI, ML, Mobile Development

CERTIFICATIONS

- Machine Learning — Stanford University
- Advanced Learning Algorithms — Stanford University
- Natural Language Processing — Stanford University
- AI For Everyone — Stanford University
- ChatGPT Prompt Engineering — DeepLearning.AI
- UX Design Foundations — Google

LANGUAGES

English (Professional) | Urdu (Native) | Pashto (Native)